MIL-A-85097/8B(AS)
24 August 1984
SUPERSEDING
MIL-A-85097/8A(AS)
26 June 1980

MILITARY SPECIFICATION SHEET

AIRCREW ESCAPE PROPULSION SYSTEMS (AEPS); ROCKET MOTORS, UNDER SEAT, GAS INITIATED (MARK 74 MOD 0, MARK 75 MOD 0, MARK 76 MOD 0, MARK 77 MOD 0, MARK 86 MOD 0, MARK 87 MOD 0, MARK 88 MOD 0, MARK 92 MOD 0, MARK 100 MOD 0, AND MARK 101 MOD 0) (LOADED ASSEMBLY)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the Gas Initiated, Under Seat Rocket Motors described herein shall consist of this document and the latest issue of MIL-A-85097.

1. SCOPE

1.1 Scope. This specification sheet covers requirements unique to ten types of gas initiated, under seat rocket motors (loaded assemblies) for use in the aircrew escape propulsion systems.

1.2 Classification. The rocket motors shall be of the following types, as specified in the contract or purchase order (see 6.2.1).

Type I	Rocket Motor Mark 74 Mod 0
Type II	Rocket Motor Mark 75 Mod 0
Type III	Rocket Motor Mark 76 Mod 0
Type IV	Rocket Motor Mark 77 Mod 0
Type V	Rocket Motor Mark 86 Mod 0
Type VI	Rocket Motor Mark 87 Mod 0
Type VII	Rocket Motor Mark 88 Mod 0
Type VIII	Rocket Motor Mark 92 Mod 0
Type IX	Rocket Motor Mark 100 Mod 0
Type X	Rocket Motor Mark 101 Mod 0

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following drawings are in addition to those listed in the cover specification, MIL-A-85097, and form a part of this specification sheet to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

DRAWINGS

Naval Air Systems Command (Code Ident 30003)

4903620	Label, Identification, Rocket Motor
4904093	Rocket Motor Mark 74 Mod 0
4904094	Rocket Motor Mark 75 Mod 0
4904095	Rocket Motor Mark 76 Mod 0
4904096	Rocket Motor Mark 77 Mod 0
4904171	Rocket Motor Mark 86 Mod 0
4904172	Rocket Motor Mark 87 Mod 0
4904173	Rocket Motor Mark 88 Mod 0
4904216	Rocket Motor Mark 92 Mod 0
1176AS200	Rocket Motor Mark 100 Mod 0
1176AS300	Rocket Motor Mark 101 Mod 0

3. REQUIREMENTS

3.1 General. The requirements in this specification sheet unless in direct conflict with the cover specification shall be considered additional, not substitute. The motors shall meet the requirements specified on the following drawings: 4903620, 4904093, 4904094, 4904095, 4904096, 4904171, 4904172, 4904173, 4904216, 1176AS200, 1176AS300.

3.2 Preproduction.

3.2.1 Propellant aging. When tested in accordance with MIL-A-85097, the propellant shall meet the limit requirements specified by the contractor and which are acceptable to the Government.

3.3 Preproduction/production.

3.3.1 Propellant. The propellant shall conform to the contractor's in-house propellant specification when tested for the data of MIL-A-85097. The contractor shall make available the results of the tested values and the in-house specification limits for comparison (see 6.2.2).

3.3.2 Life. Life shall conform to the requirement of MIL-A-85097 with the exception of propellant life which shall be 3 years maximum on sheet stock and 1 year on extruded propellant grain.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

3.3.3 Ballistic performance. Rocket Motors Mark 74/75/76/77/86/87 Mod 0 and Mark 88 Mod 0 shall meet the ballistic performance requirements specified in table I. Rocket Motor Mark 92 Mod 0 shall meet the ballistic performance requirements specified in table II. Rocket Motors Mark 100/101 Mod 0 shall meet the ballistic performance requirements specified in table III.

TABLE I. Ballistic performance requirements for Mk 74/75/76/77/86/87/88 Mod 0 Rocket Motors.

Parameter	40 ± 5°F		160° ± 5°F	
	Minimum	Maximum	Minimum	Maximum
Rocket motor ignition delay (ms)	-	17.0	-	17.0
Maximum rocket motor thrust (lbf)	4,270	4,910	5,060	6,200
Rocket motor action time (ms)	280	310	245	275
Rocket motor impulse (lbf-s)	1,200	1,290	1,225	1,315

TABLE II. Ballistic requirements for Mk 92 Mod 0 Rocket Motor.

Parameter	40 ± 5°F		160° ± 5°F	
	Minimum	Maximum	Minimum	Maximum
Rocket motor ignition delay (ms)	-	25	-	25
Maximum rocket motor thrust (lbf)	3,700	6,900	3,700	6,900
Rocket motor action time (ms)	220	475	220	475
Rocket motor impulse (lbf-s)	1,160	1,390	1,160	1,390

TABLE III. Ballistic performance requirements for Mk 100/101 Mod 0 Rocket Motors.

Parameter	40° ± 5°F		160° ± 5°F	
	Minimum	Maximum	Minimum	Maximum
Rocket motor ignition delay (ms)	-	10	-	10
Maximum rocket motor thrust (lbf)	3,520	5,280	3,520	5,280
Rocket motor action time (ms)	222	344	222	344
Rocket motor impulse (lbf-s)	976	1,464	976	1,464

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Production quality conformance inspection.

4.1.1 **Sampling.** For sampling purposes, the Rocket Motors Mk 74 Mod 0, Mk 75 Mod 0, Mk 76 Mod 0, Mk 77 Mod 0, Mk 86 Mod 0, Mk 87 Mod 0, Mk 88 Mod 0, Mk 100 Mod 0, and Mk 101 Mod 0 may be considered a single lot when loaded from the same propellant lot. Individual sample quantities of the various rocket motors comprising the total lot acceptance sample shall be proportionate to the corresponding quantity in the lot.

4.2 Examinations and tests.

4.2.1 Preproduction.

4.2.1.1 Environmental.

4.2.1.1.1 **Vibration.** For the AEPS covered by this specification sheet, the following test procedures shall be used in lieu of those specified in the general specification. Two rocket motors shall be subjected to vibration testing. The Mk 74/75/76/77/86/87/88/92/100 and 101 Mod 0 rocket motors shall be temperature conditioned for a minimum of 6 hours, one at $-40° \pm 5°F$ and the other at $+160° \pm 5°F$. The rocket motors shall be maintained at specified conditioning temperature during these tests. If the rocket motor must be removed from the conditioning chamber or if the rocket motor temperature falls outside the specified temperature limits for more than 5 minutes, the testing shall stop and the rocket motor shall be reconditioned. A reconditioning time of 2 minutes for every minute the rocket motor spent outside the specified temperature envelope is required. Reconditioning time shall be not greater than 8 hours. A continuous temperature history of the rocket motor shall be kept for all conditioning and test time. Excitation shall be applied through the aircraft seat attachment fittings unless otherwise noted in the contract (see 6.2.1). Accelerometers to measure acceleration along the axis perpendicular to the axis through the attachment points of the rocket motor shall be placed midway between the attachment points. Longitudinal vibration shall be measured at either end of the rocket motor.

4.2.1.1.1.1 **Resonance survey.** Resonance surveys shall be conducted on two loaded rocket motors in three axes at the maximum and minimum conditioning equilibrium temperatures. Input vibratory acceleration of 5 g's shall be applied successively along each of the axes noted above when sweeping between 10 and 499 hertz. Input vibratory accelerations of 3 g's shall be applied for sweeps between 500 and 2,000 hertz. All resonances shall be as noted. During the resonance survey a permanent, continuous record of input vibratory acceleration and of unit response to the input vibratory acceleration shall be kept. This record shall be made on an instrument acceptable to the Government and shall include the response along each of the three axes. Frequency shall be varied at a rate not greater than 0.25 octave per minute.

a. A resonance shall be defined as a frequency at which a clearly defined peak in the amplitude ratio (ratio of the output acceleration in any of the three igniter cartridge axes to the input acceleration as measured at the excitation attachment points) exceeds a value of 1.5 to 1.

b. If the resonance survey indicates a frequency band wherein the amplitude ratio exceeds 1.5 to 1 (as contrasted to a clearly defined peak) this entire band width shall be considered one resonant point. The minimum frequency band width which shall be considered a resonant band shall be 0.33 octave. If, also within this band width, there are clearly defined amplitude peaks, the most severe of these shall be considered and additional resonant point.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

4.4.2.4.2 Endurance vibration. Endurance vibration shall be conducted on two loaded rocket motors at the maximum and minimum conditioning equilibrium temperatures. Each rocket motor shall be vibrated for a minimum of 24 hours, 8 hours in each of the three axes. The vibration time in each axis shall be equally divided between the resonant frequencies noted in that axis during the resonant survey. A peak input acceleration of 5 g's shall be maintained at frequencies between 10 and 499 hertz and 3 g's for frequencies between 500 and 2,000 hertz. If no resonant frequency is detected in an axis, the rocket motor shall be vibrated in that axis for a minimum of 2 hours at 5 g's input through the 10 to 499 hertz range and 6 hours at 3 g's input through the 500 to 2,000 hertz range. Total vibration time for the rocket motors shall be a minimum of 24 hours which shall include resonant and nonresonant frequencies, as applicable. At the conclusion of the tests, the rocket motor shall be visually and radiographically inspected for damage without disassembly of the rocket motor. The rocket motors shall be fired at the same temperature at which they were conditioned and vibrated.

4.3 Preproduction/production.

4.3.1 Conditioning temperatures. The conditioning temperatures and tolerance shall be as follows:

a. Cold: $-40° \pm 5°F$
b. Hot: $160° \pm 5°F$

4.3.2 Conditioning period. Time for conditioning prior to static firing or during temperature cycling shall be as follows:

a. Static firing: 6 hours, minimum
b. Temperature cycling: 6 to 8 hours

5. PACKAGING

(This section is not applicable to this specification sheet.)

6. NOTES

6.1 Intended use. The rocket motors described herein are intended for use as a means of ejecting personnel from the following aircraft:

Aircraft	Rocket motor
F-14	Mk 74 Mod 0 and Mk 75 Mod 0
A-6	Mk 76 Mod 0 and Mk 77 Mod 0
EA-6B	Mk 86 Mod 0, Mk 87 Mod 0, and Mk 88 Mod 0
F-4	Mk 92 Mod 0
F/A-18	Mk 100 Mod 0
TF-18	Mk 100 Mod 0 and Mk 101 Mod 0

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Mark and Mod designation of rocket motor
c. Direction for excitation of rocket motor, if other than as specified in 4.2.1.1.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

6.2.2 **Data requirements.** When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of FAR 52-209-3 and -4 are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
3.3.1	Acceptance Test Report	UDI-T-22741A	-

(Data item descriptions related to this specification, and identified in this section will be approved and listed as such in DoD 5000.19L., Vol.II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 **Definitions.**

6.3.1 **Rocket motor ignition delay.** Rocket motor ignition delay is the elapsed time from shear pin failure to first indication of rocket motor thrust.

6.3.2 **Maximum rocket motor thrust.** The maximum rocket motor thrust is that obtained by measuring the maximum vertical rocket thrust.

6.4 **Changes from previous issue.** Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - AS
(Project 1377-N756)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-85097/8B(AS)	AIRCREW ESCAPE PROPULSION SYSTEMS (ACES II EJECTION SEAT MOTORS, UNDER SEAT, GAS INITIATED (MARK 74 ROD O....MARK 101 MOD (LOADED ASSEMBLY)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 02 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(ESSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T10:23Z
Check the source to verify that this is the current version before use.